SE Lab 6: Code and Branch Coverage

Highway Toll Calculator: Lab Report

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Assigned repo: github.com/SETAPESU26/L6_L7_36_highway_toll

My repo: github.com/VismayCD/L6_L7_36_highway_toll

1. Objective

To measure the test coverage of highway_toll.py using coverage and pytest, and to write test cases that reach at least 90% code coverage and at least 90% branch coverage.

2. Initial Coverage

The two starter tests pass but leave highway_toll.py at 39% coverage, with 15 of 28 statements missed (lines 7, 11, 13, 15-18, 25 and 29-36). Five of its nine decisions only ever went one way, and the other four never ran.

3. Code Coverage

Code coverage checks that every line runs at least once. CodeCoverageTest_597_highway_toll.py keeps the starter tests and adds tests for the lines they never reach:

Lines covered	Test case
7	Fewer than 2 axles raises ValueError
11	5 axles uses the flat rate of 400
13	Peak hours add 25%
15-18	FASTag discount: 10% for 2 axles, 5% for 3 axles
25	A non-exempt vehicle pays the normal toll
29-36	pass_advice returns buy_pass, either and pay_per_trip

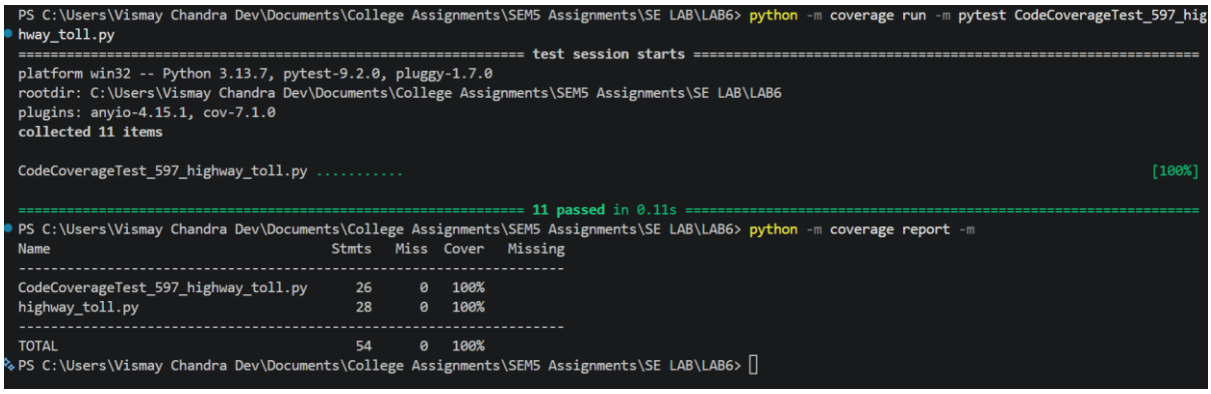


Figure 1: Code coverage. 11 tests passed, 0 of 28 statements missed, 100% coverage.

4. Branch Coverage

Branch coverage checks that every decision goes both ways. highway_toll.py has nine decisions, which gives 18 branches. BranchCoverageTest_597_highway_toll.py tests each one in both directions, including boundary values:

Decision	One way	Other way
Line 6: axles < 2	1, 0, -1 axles	2 axles
Line 8: axles in BASE_TOLL	2, 3, 4 axles	5, 6, 12 axles
Line 12: is_peak	Peak	Off-peak
Line 14: has_fastag	FASTag	No FASTag
Line 15: axles == 2	2 axles	3, 4, 5 axles
Line 23: vehicle in EXEMPT	ambulance, fire_engine	car, truck
Line 30: for ... in trips	One or more trips	No trips
Line 32: cost > pass_price	Pass price 199	Pass price 200, 201
Line 34: cost == pass_price	Pass price 200	Pass price 201

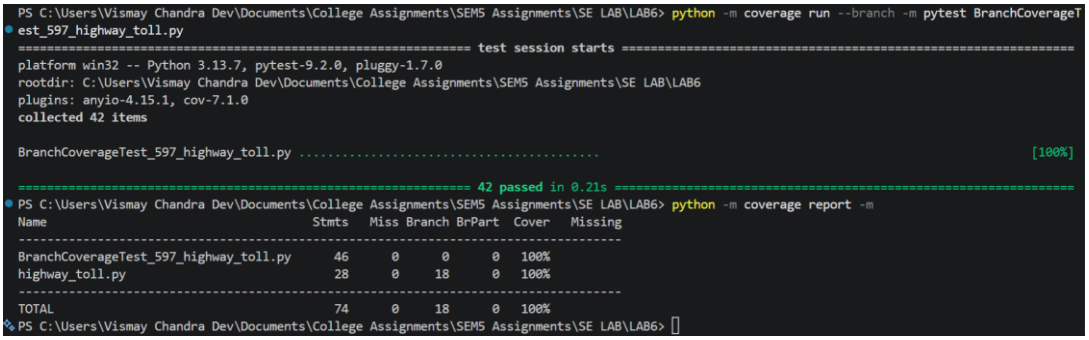


Figure 2: Branch coverage. 42 tests passed, 18 of 18 branches covered, 100% coverage.

5. Results

highway_toll.py	Starter tests	Code coverage tests	Branch coverage tests
Tests passed	2	11	42
Statements missed	15 of 28	0 of 28	0 of 28
Branches covered	5 of 18	Not measured	18 of 18
Coverage reported	39%	100%	100%

Both targets of at least 90% were met.